BDA400
DATA SCIENCE TOOLS AND TECHNIQUES

Assignment 2
Technical Analysis Using R
Preliminary Stage

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Assessment weight: 5%

Assignment report and implementation guide

Prepared: October 2026

Submission status: DRAFT — complete the clearly marked verification and screenshot items before submitting.

GitHub repository and deliverables

Repository name required by the instructor: TechnicalAnalysis

Repository link: [INSERT THE ACTUAL REPOSITORY URL OWNED/USED FOR HANIN'S SUBMISSION — NOT VERIFIED]

Recommended folder: TechnicalAnalysis/A02_Preliminary_Stage/. Upload this report, portfolio.txt, technical_analysis.R, and actual output/screenshots after execution.

Files supplied alongside this report: portfolio.txt, technical_analysis.R, and README.md. The R script creates stock_statistics.csv and one PNG chart per symbol after successful execution.

1. Executive summary and objectives

This preliminary stage establishes a reproducible stock-data analysis workflow in R. A text-based portfolio identifies the securities, quantmod obtains historical market data, descriptive statistics summarize closing prices, and a 20-session simple moving average supports basic trend exploration. Results are displayed as tabular output and price charts. The design separates data retrieval, statistics, presentation, and execution, making the workflow easier to test and extend in later assignments.

The analysis is descriptive and educational. It does not forecast returns or constitute investment advice. No claim is made that installation, package execution, data downloads, or image capture has occurred until accompanied by the student's genuine evidence.

2. Project structure and workflow

Item	Purpose
TechnicalAnalysis repository	Version-controlled record of course assignments
A02_Preliminary_Stage/portfolio.txt	Input file: one ticker per line
A02_Preliminary_Stage/technical_analysis.R	Import, calculations, summaries and visualization
A02_Preliminary_Stage/stock_statistics.csv	Generated descriptive summary after execution
A02_Preliminary_Stage/charts/*.png	Generated visualization after execution
A02_Preliminary_Stage/report	Step-by-step report with genuine screenshots

Processing pipeline: portfolio.txt → read_portfolio() → load_stock_data() → calculate_statistics() → show_stock_data() → plot_stock_data() → CSV + PNG outputs.

3. GitHub setup and repository configuration (10 points)

Procedure: (1) Sign in to GitHub. (2) Create a repository named TechnicalAnalysis, or confirm an existing appropriate repository. (3) Create an A02_Preliminary_Stage subfolder. (4) Commit portfolio.txt, technical_analysis.R, the report, and README. (5) Check that files open correctly and copy the repository URL into the cover section.

SCREENSHOT REQUIRED — Figure 1 — GitHub repository
Show repository name, account, URL and the A02 folder.
[Insert authentic screenshot from Hanin's PC / GitHub here]

4. R and RStudio installation (15 points)

Install the appropriate R build from CRAN (https://cran.r-project.org/), then install RStudio Desktop from Posit. Open RStudio and use the Console to run R.version.string. Record the installed version and confirm that the Source pane, Console, Environment pane and Files/Plots pane are visible. Screenshots of actual installation/setup are important for the grading rubric.

SCREENSHOT REQUIRED — Figure 2 — R installation
Provide an actual installer or installed-version screen; do not fabricate past installer screens.
[Insert authentic screenshot from Hanin's PC / GitHub here]

SCREENSHOT REQUIRED — Figure 3 — RStudio setup
Show RStudio opened with the Console and R version visible.
[Insert authentic screenshot from Hanin's PC / GitHub here]

5. Installing technical-analysis packages (15 points)

In the RStudio Console, execute the following commands. quantmod handles stock downloads and time-series data; TTR provides technical indicators such as the simple moving average. Dependencies are handled by install.packages().

```
install.packages(c("quantmod", "TTR"))
library(quantmod)
library(TTR)
packageVersion("quantmod")
packageVersion("TTR")
```

SCREENSHOT REQUIRED — Figure 4 — Package installation
Show the install.packages commands, successful console output, and/or packageVersion results.
[Insert authentic screenshot from Hanin's PC / GitHub here]

6. Project initialization and portfolio definition (5 points)

Create portfolio.txt using a UTF-8 plain-text editor. The provided file uses three example US-listed stocks, one per line. Tickers can be replaced with the student's chosen securities before execution.

```
AAPL
MSFT
GOOGL
```

At execution, read_portfolio() removes whitespace, ignores empty lines and comments, and de-duplicates tickers. The script expects portfolio.txt in the current R working directory.

SCREENSHOT REQUIRED — Figure 5 — Portfolio configuration
Show portfolio.txt opened in RStudio and the project files in the Files pane.
[Insert authentic screenshot from Hanin's PC / GitHub here]

7. Stock-data import function (15 points)

The load_stock_data(symbols, from, to) function iterates through the ticker vector, downloads approximately one year of daily OHLCV data with quantmod::getSymbols(..., src="yahoo", auto.assign=FALSE), then stores the results separately by ticker in a named list of xts objects. Each object can be converted to a data frame for inspection. Failed tickers raise warnings instead of silently contaminating valid results.

Inputs: stock ticker vector, start date and end date. Output: a named list keyed by the stock symbol. The retrieval requires a working Internet connection and availability of the external data source.

SCREENSHOT REQUIRED — Figure 6 — Imported stock data
Show successful Console loading messages and sample OHLCV rows for all downloaded tickers.
[Insert authentic screenshot from Hanin's PC / GitHub here]

8. Descriptive statistics and methods (20 points)

The calculate_statistics() function extracts daily closing prices and calculates a separate summary for every successfully loaded security. Missing/non-finite closes are removed prior to computation.

Measure	Definition / interpretation
Latest closing price	Most recent available daily close in the chosen sample
Mean	Arithmetic average of available daily closes
Median	Middle observation after prices are ordered
Mode (rounded cents)	Most frequent 2-decimal close; smallest value chosen if tied
Standard deviation	Sample SD of close prices; price variability in USD
SMA(20)	Mean of the most recent 20 daily closing prices
Observations	Count of finite close-price records

A statistical caution: share prices are approximately continuous, so an exact mode often has little meaning. For reproducibility, the assignment reports the most frequent close rounded to two decimals and discloses this operational definition. The SMA(20) is NA where the sample has fewer than 20 observations. Standard deviation measures historical price dispersion, not necessarily financial risk or future volatility.

9. Output display and visualizations (10 points)

show_stock_data() prints the latest six OHLCV records for each ticker. calculate_statistics() returns a structured summary data frame, exported to stock_statistics.csv. plot_stock_data() generates a labeled PNG line chart for each ticker containing the closing-price line and a 20-session SMA line, with legend, date axis, and price axis.

SCREENSHOT REQUIRED — Figure 7 — Statistical summary
Show genuine console data-frame output and/or the generated stock_statistics.csv.
[Insert authentic screenshot from Hanin's PC / GitHub here]

SCREENSHOT REQUIRED — Figure 8 — Price and SMA charts
Include at least one actual PNG chart produced by the script; ideally show all portfolio symbols.
[Insert authentic screenshot from Hanin's PC / GitHub here]

10. Reproducibility, validation and limitations

- Execute in RStudio with a stable Internet connection and portfolio.txt in the current working directory.
- Verify that the number of loaded tickers matches expected symbols; investigate warning messages.
- Check that each output summary has a positive observation count and that the final close matches the raw series.
- Validate mean, median and SD for one ticker using direct base-R calculations on its closing prices.
- Check that each SMA(20) begins only after enough trading sessions, and is plotted over the correct ticker.
- Daily downloaded prices and last available trading date vary with the execution date and external data provider.
- Missing observations, corporate actions, ticker changes and third-party data interruptions can affect interpretation.

SCREENSHOT REQUIRED — Figure 9 — Verification evidence
Add a console screenshot showing a manual check of one statistic and the created output files.
[Insert authentic screenshot from Hanin's PC / GitHub here]

11. Results and reflection — complete after execution

Execution date: [INSERT] R version: [INSERT] quantmod version: [INSERT] TTR version: [INSERT]

Tickers loaded successfully: [INSERT]. Date range actually returned: [INSERT]. Any failed ticker(s): [INSERT / None].

Observed results: [INSERT one or two sentences discussing the actual close-price trends and SMA relationships seen in your generated charts.]

Reflection: The project demonstrates how a simple text input and modular R functions form a reusable analytical pipeline. Separating loading, summary statistics and visualization improves traceability and helps pinpoint errors. Data quality checks and careful interpretation are essential because market prices are dynamic and descriptive statistics alone cannot establish a trading signal.

12. Submission checklist

[ ] Repository TechnicalAnalysis exists and contains A02 files

[ ] R and RStudio genuinely installed and functioning

[ ] quantmod and TTR installed; versions documented

[ ] portfolio.txt contains chosen symbols

[ ] Script run successfully; no unexplained errors

[ ] stock_statistics.csv and chart PNGs generated

[ ] Figures 1–9 replaced with genuine screenshots

[ ] Actual results and repository URL filled in

[ ] Report re-exported as correctly named PDF

[ ] Relevant assignment files committed to GitHub

References

CRAN: https://cran.r-project.org/
Posit RStudio: https://posit.co/download/rstudio-desktop/
GitHub documentation: https://docs.github.com/en/get-started
quantmod package: https://cran.r-project.org/package=quantmod
TTR package: https://cran.r-project.org/package=TTR
Yahoo Finance (market data source): https://finance.yahoo.com/